Relations, Products, and the First Appearance of Universal Properties

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§1.1 Relations and equivalence classes

This note continues the foundational setup, moving from sets and functions to relations, products, and the integers. The original title says "Intergers"; I will silently correct this to *integers*.

Definition 1.1.1. For sets or classes A and B, their Cartesian product is

$$A \times B = \{(a, b) \mid a \in A,\ b \in B\}.$$

A **relation from A to B** is a subclass of $A \times B$. A **relation on A** is a subclass of $A \times A$.

Caution

The original note has the phrase "a subset of R is a relation on $A \times B$", which is clearly a slip. The intended statement is that a relation is a subset of $A \times B$.

Definition 1.1.2. A relation $R \subseteq A \times A$ is an **equivalence relation** if it is:

(i) reflexive: $(a, a) \in R$ for all $a \in A$;

(ii) symmetric: $(a, b) \in R$ implies $(b, a) \in R$;

(iii) transitive: $(a, b), (b, c) \in R$ implies $(a, c) \in R$.

When R is an equivalence relation, we write $a \sim b$ for $(a, b) \in R$, and define the equivalence class of a by

$$[a] = \{x \in A \mid x \sim a\}.$$

The quotient class is

$$A/R = \{[a] \mid a \in A\}.$$

Proposition 1.1.3

If R is an equivalence relation on A, then:

(i) $[a] \neq \varnothing$ for every $a \in A$;

(ii) $A = \bigcup_{a \in A} [a] = \bigcup_{C \in A/R} C$;

(iii) $[a] = [b]$ if and only if $a \sim b$.

Lemma 1.1.4

For any $a, b \in A$, either $[a] \cap [b] = \varnothing$, or $[a] = [b]$.

Proof. If $x \in [a] \cap [b]$, then $x \sim a$ and $x \sim b$. By symmetry and transitivity, $a \sim b$, and hence any element equivalent to a is equivalent to b, and conversely. Thus $[a] = [b]$. $\square$

§1.2 Partitions

Definition 1.2.1. Let A be a nonempty class. A family $\{A_i \mid i \in I\}$ of subclasses of A is a **partition** of A if

(i) $A_i \neq \varnothing$ for every $i \in I$;

(ii) $\bigcup_{i \in I} A_i = A$;

(iii) $A_i \cap A_j = \varnothing$ whenever $i \neq j$.

> **Remark 1.2.2** — This is one of the most useful ways to remember equivalence relations: an equivalence relation cuts A into disjoint blocks, and those blocks form a partition. Conversely, a partition defines an equivalence relation by declaring two elements equivalent exactly when they lie in the same block.

§1.3 Indexed products and projections

The note then generalizes finite Cartesian products. For a set-indexed family $\{A_i \mid i \in I\}$, define

$$\prod_{i \in I} A_i$$

to be the set of functions

$$f : I \to \bigcup_{i \in I} A_i$$

such that $f(i) \in A_i$ for every $i \in I$.

When $I = \{1, 2, \dots, n\}$, this agrees with the usual product

$$A_1 \times A_2 \times \cdots \times A_n,$$

because such a function is the same information as the ordered n-tuple

$$(f(1), f(2), \dots, f(n)).$$

> **Remark 1.3.1** — The original note pauses over the case $I = \{1, 2\}$. A point of $A_1 \times A_2$ may be written as a pair (a_1, a_2), but it may also be regarded as a function f with $f(1) = a_1$ and $f(2) = a_2$. This is a small but important shift: products are not just lists; they are choice-like functions over an index set.

For each $k \in I$, the **canonical projection**

$$\pi_k : \prod_{i \in I} A_i \to A_k$$

is defined by

$$\pi_k(f) = f(k).$$

Example 1.3.2
If $I = \mathbb{R}$ and $A_i = \mathbb{C}$ for all i, then an element of $\prod_{i \in I} A_i$ is a function $f : \mathbb{R} \to \mathbb{C}$. For example, $f(x) = ix$. The projection at $k \in \mathbb{R}$ returns

$$\pi_k(f) = f(k) = ik.$$

§1.4 The first universal property

Here the original note becomes excited: "Appears! Universal property!" This is worth preserving. Products are not only sets of tuples; they are characterized by how maps into them behave.

Theorem 1.4.1 (Universal property of products)
Let $\{A_i \mid i \in I\}$ be a family of sets. There is a set D, equipped with maps

$$\pi_i : D \to A_i \qquad (i \in I),$$

such that for every set C and every family of maps $f_i : C \to A_i$, there is a unique map

$$f : C \to D$$

with

$$\pi_i \circ f = f_i$$

for every $i \in I$. Moreover, D is unique up to a unique bijection compatible with the projections.

Proof. Take $D = \prod_{i \in I} A_i$. Given maps $f_i : C \to A_i$, define

$$f(c)(i) = f_i(c).$$

Then $f(c) \in \prod_i A_i$, and $\pi_i(f(c)) = f_i(c)$. Uniqueness follows because a point of the product is determined by all of its coordinates. □

Remark 1.4.2 — The original note says that the proof is a little long and should maybe be postponed until systematic study of category theory. The cleaned-up proof above is short because we already know the concrete product. The deeper point is that the same pattern will later define products in arbitrary categories, where elements and coordinates may no longer be available.

§1.5 Integers and induction

The note then turns to the natural numbers and the integers through Peano's axioms. In modern notation the Peano axioms say, roughly:

(i) 0 is a natural number;

(ii) every natural number has a successor;

(iii) 0 is not the successor of any natural number;

(iv) distinct natural numbers have distinct successors;

(v) any property true of 0 and inherited by successors is true of all natural numbers.

The fifth axiom is the principle of mathematical induction.

Theorem 1.5.1 (Induction and strong induction)
Let $S \subseteq \mathbb{N}$ with $0 \in S$. If either

(i) $n \in S$ implies $n+1 \in S$ for all $n \in \mathbb{N}$, or

(ii) $\{0, 1, \ldots, n-1\} \subseteq S$ implies $n \in S$ for all $n \in \mathbb{N}$,

then $S = \mathbb{N}$.

Caution

The original note proves this by taking the least element of $\mathbb{N} \setminus S$. That proof is clean if the well-ordering principle for $\mathbb{N}$ is already available. If one is deriving the theory strictly from Peano's axioms, one should first justify the well-ordering principle or prove the statement directly by induction.

§1.6 Division, gcd, and congruence

Theorem 1.6.1 (Division algorithm)
If $a, b \in \mathbb{Z}$ and $a \neq 0$, then there exist unique integers q, r such that

$$b = aq + r, \qquad 0 \leq r < |a|.$$

Definition 1.6.2. For $a \neq 0$, we write $a \mid b$ if there exists $k \in \mathbb{Z}$ such that $ak = b$. Otherwise we write $a \nmid b$.

Definition 1.6.3. The positive integer c is the greatest common divisor of $a_1, a_2, \ldots, a_n$ if

(i) $c \mid a_i$ for every $1 \leq i \leq n$;

(ii) whenever $d \in \mathbb{Z}$ and $d \mid a_i$ for every i, then $d \mid c$.

It is denoted $(a_1, a_2, \ldots, a_n)$.

Definition 1.6.4. Let $m > 0$. If $a, b \in \mathbb{Z}$ and $m \mid (a - b)$, then a is congruent to b modulo m, written

$$a \equiv b \pmod{m}.$$

Theorem 1.6.5

Let $m > 0$ and $a, b, c, d \in \mathbb{Z}$.

(i) Congruence modulo m is an equivalence relation on $\mathbb{Z}$, with precisely m equivalence classes.

(ii) If $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$, then

$$a + c \equiv b + d \pmod{m}, \qquad ac \equiv bd \pmod{m}.$$

(iii) If $ab \equiv ac \pmod{m}$ and $(a, m) = 1$, then $b \equiv c \pmod{m}$.

Proof. The original note leaves this as an exercise. The key point is divisibility. For example,

$$m \mid (a - b), \qquad m \mid (c - d)$$

imply

$$m \mid (a + c) - (b + d).$$

For multiplication, write

$$ac - bd = c(a - b) + b(c - d).$$

For cancellation, $ab \equiv ac \pmod{m}$ means $m \mid a(b - c)$. If $(a, m) = 1$, Euclid's lemma gives $m \mid (b - c)$, hence $b \equiv c \pmod{m}$. $\square$